Atoms & nuclear equations

Count & explain

1 Count particles

Proton +1, neutron 0, electron −1. Neutral: protons = electrons.

Carbon-14: 6p, 8n, 6e. Isotopes share proton number; ions differ in electron count.

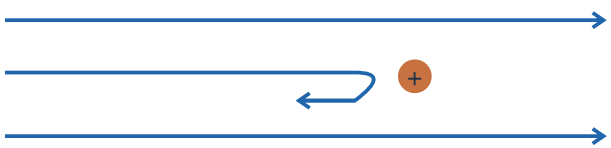
2 Size & levels

Atom radius ≈ 10^{-10} m; nucleus less than 1/10,000 of that, holding almost all mass.

Absorb radiation → higher electron level. Emit → lower. Removing an electron ionises the atom.

Follow evidence

3 Alpha scattering



Most pass; very few turn back

Most pass straight → mostly empty space. Some deflect → positive repulsion.

Very few return → mass and charge concentrated in tiny nucleus; plum pudding fails.

4 Models develop

Indivisible sphere → electron discovery → plum pudding → nuclear atom.

Bohr adds specific electron levels; later proton identification; Chadwick gives neutron evidence.

Balance decays

5 Alpha & beta-minus

Alpha is $^{4}_{2}\mathrm{He}$: daughter **A−4, Z−2.**

Beta-minus: neutron → proton + emitted electron; daughter **A unchanged, Z+1.**

Electron is from nucleus, not shell.

6 Balance both numbers

$^{238}_{92}\mathrm{U} \rightarrow {}^{234}_{90}\mathrm{Th} + {}^{4}_{2}\mathrm{He}$.

$^{14}_{6}\mathrm{C} \rightarrow {}^{14}_{7}\mathrm{N} + {}^{0}_{-1}\mathrm{e}$. Totals of A and Z match on both sides.

Gamma loses energy with no change in A or Z.